

POPULATION BIOLOGY AND SPATIAL RELATIONSHIPS OF COEXISTING SPINY MICE (*ACOMYS*) IN ISRAEL

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Nocturnally active *Acomys cahirinus* and diurnally active *A. russatus* cooccur in hot rocky deserts, and their temporal partitioning results from competition. We studied their population biology at Ein Gedi near the Dead Sea to determine their spatial relationships and gain insight into their ecological overlap. Individuals of both species were trapped repeatedly for almost 2 years. Mean body mass did not change seasonally, and *A. russatus* was significantly heavier than *A. cahirinus*. Density of *A. russatus* was significantly greater than that of *A. cahirinus*. *Acomys russatus* had a shorter reproductive peak that overlapped the longer reproductive season of *A. cahirinus*. *Acomys russatus* showed a significant preference for boulder habitat versus open habitat at all seasons, whereas *A. cahirinus* showed a preference for boulder habitat only half of the time. Sexes of both species overlapped in home range.

Key words: *Acomys*, body mass, microhabitat choice, population biology, spiny mice, temporal partitioning

Two species of spiny mice (*Acomys*) coexist on rocky terrain in southern Israel. The common spiny mouse (*A. cahirinus*) is geographically more widespread, with populations ranging throughout North Africa (including the Sahara), East Africa, Turkey, Iran, and Pakistan (Harrison and Bates 1991). In Israel, it is found in all rocky areas and some nonrocky areas such as the Jordan and Arava valleys (Mendelssohn and Yom-Tov 1987). The golden spiny mouse (*A. russatus*) is found in deserts ranging from eastern Egypt through the Sinai Peninsula to the Arabian Peninsula (Harrison and Bates 1991). In Israel, it is restricted to rocky deserts south and east of the 100-mm isohyet (Mendelssohn and Yom-Tov 1987).

A striking feature of these 2 species is their temporal segregation; the common spiny mouse is nocturnal, whereas the golden spiny mouse is diurnal (Qumsiyeh 1996). Such definitive temporal segregation

is unusual among congeners because major morphological, physiological, and behavioral adaptations often accompany specialization to nocturnal and diurnal ways of life. Therefore, congeneric and to a large extent even confamilial species generally are active during the same part of the diel cycle (Daan 1981). Shkolnik (1971) discovered that when the common spiny mouse was removed from a shared habitat, the golden spiny mouse became nocturnal. The shift in times of activity of the golden spiny mouse in absence of its congener implies that the 2 species compete (Shkolnik 1971).

Understanding population ecology of the 2 species of spiny mouse and their spatial relationships at the level of local coexistence is crucial for understanding their ecological relationship. Although mechanisms enabling coexistence and structuring of ecological communities have received considerable attention in recent years in sandy-desert rodent communities (Brown 1987; Kotler and Brown 1988), little is known about rocky-desert rodent communities.

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We carried out a 25-month research project on coexisting populations of golden spiny mice and common spiny mice at Ein Gedi, near the Dead Sea, to determine their spatial relationships and gain further insight into the ecological overlap between them. We obtained data on population densities, longevity, reproduction, body mass, spatial relationships, and microhabitat use in these 2 species.

MATERIALS AND METHODS

We conducted our research in the vicinity of the Ein Gedi Nature Reserve (31°28'N, 35°23'E, 100–350 m below mean sea level). Ein Gedi is characterized by steep rocky escarpments, deep canyons, and sheer dolomite cliffs that run parallel to the Dead Sea. Our study site was located just west of Kibbutz Ein Gedi on a rocky slope bisected by a wadi (dry creek bed) that was strewn with large boulders (10% of the research area). The slope is divided between an area covered with smaller stones and rocks (open habitat, 58%) and an area covered with large boulders (32%). Our trapping grid covered about 5,000 m² (0.5 ha) and included all 3 habitat types (open, boulders, and creek).

We trapped spiny mice for 25 consecutive months by setting 200 collapsible Sherman traps during the 1st year and 100 traps thereafter. Trapping sessions always were carried out during the half-moon phase. Traps were set in 5 rows that ran along the hillside. Rows were set about 5 m apart, and traps in each row also were set at 5-m intervals the 1st year and at 10-m intervals during the 2nd year. Mice were trapped every month for 3 consecutive days and nights.

Each individual captured was weighed (with a Pesola dynamometer scale to the nearest 2 g), its reproductive state was determined (scrotal testes for males and pregnancy and evident nipples in females), and it was marked by individual toe clipping. Young individuals were recognized by their weight: <30 g for *A. cahirinus* and <35 g for *A. russatus* at 1st capture (Mann 1986). We checked to see that they actually did increase in weight later, so as not to confound young mice with mice that are exceptionally small or in poor physiological condition.

Densities of spiny mice were estimated with the Jolly-Seber method (Krebs 1989). Mean densities of the 2 species were compared using

Student's *t*-test. Trappability was calculated for both species as the number of times that a mouse was caught minus 2 divided by the number of times it was exposed to capture, expressed as a percentage (Taitt 1981). We tested trappability for each sex of each species separately and compared results using the Kruskal–Wallis test.

We calculated the expected survivorship of the 2 species (Krebs 1989) and compared mean survivorships of each throughout the year using Student's *t*-test. The deviation of sex ratios from 1:1 within species was tested with a binomial test (Zar 1996). Reproductive season was determined on the basis of occurrence of males and females in reproductive condition.

Most individuals were trapped repeatedly, so we calculated mean body mass per individual for each 3-day trapping session and then calculated the mean for all individuals trapped during each month. Because body-mass data were not normally distributed (Shapiro–Wilks *W*-test), we used a Mann–Whitney *U*-test (Sokal and Rohlf 1995) to study body-mass differences between sexes and species.

Home ranges were determined (using Wildtrak, version 1.22, a nonparametric home-range analysis program) by the minimum convex polygon method, using 90% of the trap positions as locations. We used data from individuals that were caught ≥ 7 times during the study period. Although use of data from only 7 captures may result in underestimation of home-range size, that procedure was unavoidable because few individuals were caught > 7 times. We compared home-range sizes between sexes and species using a Kruskal–Wallis test (Sokal and Rohlf 1995).

We pooled trapping data of all seasons (3 months) and used chi-square tests for goodness of fit (Sokal and Rohlf 1995) to compare observed frequencies of captures with those expected based on distributions of traps in different microhabitats. In cases where we had too few data, we ignored the smallest microhabitat (wadi) and analyzed only the 2 other microhabitats (boulder and open). To detect differences in microhabitat preferences between the 2 species, we used a *G*-test (Sokal and Rohlf 1995). For each individual, only 1 capture in a specific trap during a trapping session was used in the analyses.

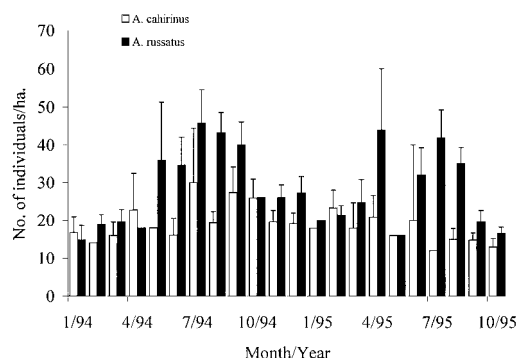


FIG. 1.—Monthly density (number of individuals per hectare) estimates ($\bar{X}$ and SE) of *A. cahirinus* and *A. russatus* calculated using the Jolly–Seber method.

RESULTS

Individuals of both species were easily and repeatedly trapped in our study site. One male *A. russatus* was trapped for 24 consecutive months, and another individual of that species and 2 individual *A. cahirinus* were trapped for ≥ 19 consecutive months. We also trapped 14 individuals of the bushy-tailed jird, *Sekeetamys calurus* (Shargal et al. 1998).

For both species of *Acomys*, we trapped more males than females. We marked 27 males and 16 females of *A. russatus* and 26 males and 16 females of *A. cahirinus*. For *A. russatus*, the difference from an expected sex ratio of 1:1 was significant ($Z = 1.68$, $P = 0.046$), whereas the difference for *A. cahirinus* was only suggestive ($Z = 1.54$, $P = 0.062$).

The density estimate for *A. cahirinus* at 19 individuals/ha (range, 12–27 individuals/ha) was lower ($P < 0.001$) than that for *A. russatus* at 28 individuals/ha (range, 15–46 individuals/ha; Fig. 1). We found no differences in trappability between sexes or species. Mean trappability was 51% ($SD = 0.219$), enhancing the reliability of the Jolly–Seber results (Hilborn et al. 1976).

Mean body masses of male ($n = 77$) and female ($n = 20$) *A. cahirinus* were $35.2 \text{ g} \pm 0.73 \text{ SE}$ and $32.0 \pm 1.31 \text{ g}$, respectively. Body masses of *A. russatus* male ($n = 119$)

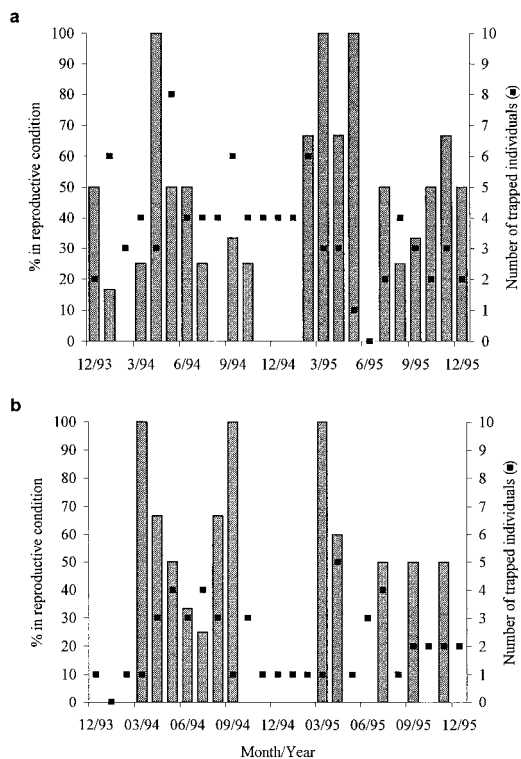


FIG. 2.—Monthly percentages of a) male and b) female *A. cahirinus* in reproductive condition and number of trapped individuals during each month.

and female ($n = 44$) were $44.1 \pm 0.61 \text{ g}$ and $41.0 \pm 1.09 \text{ g}$, respectively. Mean mass for both sexes of *A. russatus* was greater than mean mass for the same sex of *A. cahirinus* ($P < 0.001$). Males of both species were heavier than females ($P < 0.05$). No seasonal fluctuations in body mass were detected in either species.

Male *A. cahirinus* were reproductively active (showing scrotal testes) most of the year, except winter (November–January), with a peak in spring (February–May; Fig. 2a). Female *A. cahirinus* were reproductively active (pregnant and with evident nipples) between March and November (Fig. 2b). Male *A. russatus* were reproductively active beginning in January and ending in October, with a peak in January–February (Fig. 3a). Female *A. russatus* appeared reproductively active starting in

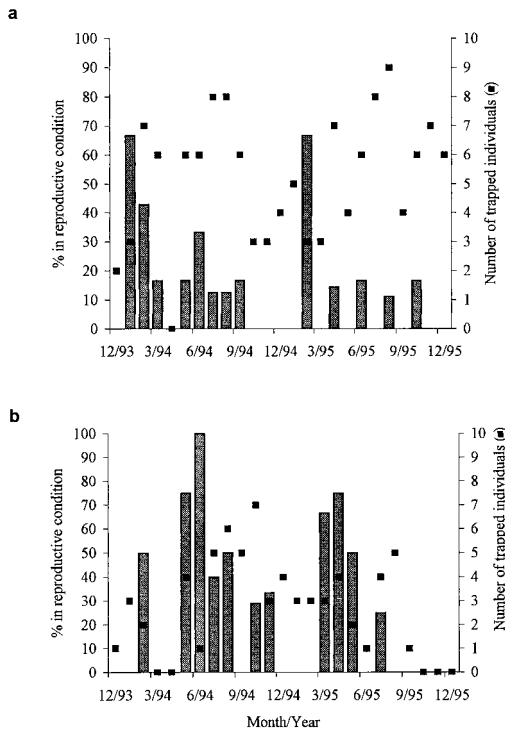


FIG. 3.—Monthly percentages of a) male and b) female *A. russatus* in reproductive condition and number of trapped individuals during each month.

February and ending in November, with a peak in March–June (Fig. 3b). Young of *A. cahirinus* appeared in May–September 1994 and February–August 1995. Young of

A. russatus appeared in May–July 1994 and April–July 1995.

In both species, mean home-range sizes of males (*A. cahirinus*, $918 \pm 212 \text{ m}^2$, $n = 7$; *A. russatus*, $533 \pm 193 \text{ m}^2$, $n = 9$) did not differ from those of females (*A. cahirinus*, $626 \pm 472 \text{ m}^2$, $n = 3$; *A. russatus*, $668 \pm 182 \text{ m}^2$, $n = 5$). We found overlap between individuals of both species and both sexes. However, because data were collected for 25 months and there were insufficient data for seasonal analysis, that result was suggestive rather than conclusive.

Individuals of *A. russatus* tended to occur in the boulder habitat throughout the year (Table 1). Individuals of *A. cahirinus* tended to occur in the boulder habitat only 50% of the time (with no evident seasonal pattern) and showed no such tendency the rest of the time (Table 1). There was a significant difference in microhabitat use between the 2 species for every 3-month period studied (Table 1).

DISCUSSION

At our study site under natural conditions in a severe hot desert, *A. russatus* has significantly higher densities than *A. cahirinus*. Previous researchers (Kam and Degen 1993; Shkolnik 1966, 1971; Shkolnik and Borut 1969) have demonstrated that *A. russatus* is physiologically more desert adapted

TABLE 1.—Number of *Acomys cahirinus* and *A. russatus* captured and comparison of intra- and interspecific habitat use (O = open habitat; B = boulder habitat; W = wadi bed).

Date	<i>A. cahirinus</i>				<i>A. russatus</i>				Interspecific comparison	
	O	B	W	P^a	O	B	W	P^a	G^b	P^b
Dec. 1993–Feb. 1994	7	18	3	<0.001	5	18	20	<0.001	10.90	<0.001
Mar. 1994–May 1994	19	26	1	<0.005	3	17	8	<0.001	16.24	<0.001
Jun. 1994–Aug. 1994	16	9	9	>0.05	10	52	18	<0.001	18.54	<0.001
Sep. 1994–Nov. 1994	14	21	4	<0.005	2	37	15	<0.001	18.55	<0.001
Dec. 1994–Feb. 1995	17	12	3	>0.05	5	46	6	<0.001	21.60	<0.001
Mar. 1995–May 1995	13	3	3	>0.05	6	21	10	<0.001	15.35	<0.001
Jun. 1995–Aug. 1995	12	3	1	>0.05	13	34	19	<0.001	17.14	<0.001
Sep. 1995–Nov. 1995	8	15	2	<0.005	4	49	14	<0.001	10.39	<0.001

^a Comparison of intraspecific habitat use within each season.

^b Comparison of interspecific differences in habitat use within each season ($d.f. = 2$ in all periods).

than is *A. cahirinus*. In contrast, in areas where food is abundant (agricultural areas and the vicinity of human settlements), densities of *A. cahirinus* are much higher than those of *A. russatus* (Shargal and Kronfeld, in litt.; Shkolnik 1966).

The breeding seasons of the 2 species of spiny mouse, based on the occurrence of reproductive characteristics, appear to overlap, but occurrence of young golden spiny mice is more restricted (April–July) than that of common spiny mice (February–September). Qumsiyeh (1996) reviewed anecdotal information on the breeding seasons of the 2 species in the Middle East and suggested that *A. cahirinus* breeds all year, whereas *A. russatus* reproduces in late spring and early summer. Densities of *A. russatus* are higher in spring and summer during their reproductive season. We detected no peak in population size for *A. cahirinus*, possibly because of their more extended period of breeding. Overlap in reproductive periods suggests that the 2 species also overlap in their peak energetic requirements for reproduction.

We found no seasonal pattern in body mass in either sex of either species. Our results conform with those of Khokhlova et al. (1994) for *A. cahirinus* at the Ramon erosion crater in the Negev Desert and contrast with their results for granivorous desert rodents (gerbillids). Among gerbillids, there were seasonal differences in body mass, with maxima in spring when food resources were relatively abundant and minima (reduced by 9–20%) in winter when food resources were relatively scarce (Brand and Abramsky 1987; Khokhlova et al. 1994). Seasonal changes in body mass often are regarded as an adaptation to reduce energy requirements or as a result of reduced food availability (Degen 1997). Granivorous desert rodents subsist on seeds of annual plants, availability of which peaks in spring, whereas food availability for omnivores probably peaks earlier during winter. Moreover, female spiny mice of both species conceive in winter and reproduce in

spring, so loss of body mass probably is not a feasible thermoregulatory mechanism in these species. This fact may account for the difference in body mass patterns between spiny mice and granivorous rodents, and, if this is the case, thermoregulation may not be the decisive factor for body mass fluctuations in desert rodents.

Mean body mass for both sexes of *A. russatus* was significantly greater than mean body mass for the same sex of *A. cahirinus*. There was no significant interspecific difference in body length for males ($P = 0.29$), but female golden spiny mice were significantly longer than female common spiny mice ($P = 0.0117$ —body length data from Mendelssohn and Yom-Tov 1987). The smaller common spiny mouse competitively displaces the larger golden spiny mouse, contrary to intuition and to most published studies of mammals (e.g., Bleich and Price 1995; Dickman 1988; Rose and Birney 1985). Although cues for temporal partitioning among spiny mice have been explored in the past few years (Haim and Rosenfeld 1993), the effect of a smaller species on a larger species remains to be studied.

It has been suggested that *A. cahirinus* is a social species (Abramsky et al. 1985). In our captive colony, both species tended to aggregate and nest in large numbers in communal nesting boxes, leaving other nesting boxes empty. The overlap in home range may indeed attest to low interindividual aggression and some pattern of social interaction within these species.

The wider and apparently more generalized distribution of common spiny mice suggests that their microhabitat requirements are less specialized than are those of golden spiny mice. Our results at the scale of local coexistence are congruent with the broad geographical patterns; common spiny mice are more widespread and less specialized also at this scale. Golden spiny mice tended to occur in the boulder microhabitat at all times, whereas common spiny mice tended to use such habitat only half of the

time, with no evident seasonal pattern. Moreover, the 2 species differed significantly in their use of microhabitats.

Microhabitat partitioning generally is considered the most significant mechanism of coexistence acting among rodent species (Morris 1987; Rosenzweig 1987). Golden spiny mice appear to be habitat specialists, whereas common spiny mice are more generalized in their habitat requirements. Although microhabitat partitioning may contribute to coexistence among spiny mice, different activity patterns may promote different microhabitat use because of alternative ecological or physiological pressures. Diurnally active golden spiny mice avoid the intensive sun of midday in summer and are active primarily during the cooler hours of morning and afternoon (Kronfeld 1998; Kronfeld et al. 1994; Shkolnik 1966). Their tendency to occur in the boulder microhabitat possibly reflects a preference for shade in this hot desert. Moreover, because different predators are active during the day and night, activity in the boulder and the open microhabitats may be accompanied by different predation threats for golden spiny mice and common spiny mice. Indirect evidence for the effect of predation consists of spines on spiny mouse rumps, a histological mechanism for tail loss (both are more marked in *A. russatus*—Shargal et al. 1999), relative immunity to the venom of the saw-scaled viper, *Echis coloratus* (studied only in *A. cahirinus*—Weissenberg et al. 1997), and a reduction of foraging by *A. cahirinus* in response to moonlight (Mandelik 1999). Thus, mechanisms driving habitat specificity in this system probably are complex.

Although the striking pattern of temporal partitioning between the 2 spiny mouse species has attracted considerable attention, field and laboratory research on spiny mice has been largely physiological (Degen 1994; Degen et al. 1986; Haim and Borut 1978) rather than ecological. Ecological research on the role of temporal partitioning for coexistence (Kronfeld and Dayan 1999)

and its physiological implications (Elvert et al. 1999; Kronfeld et al. 2000) and the role of predation in structuring this rocky desert community (Eilam et al. 1999; Jones and Dayan 2000; Mandelik 1999) are based upon our understanding of the population biology and spatial relationships of these coexisting spiny mice.

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